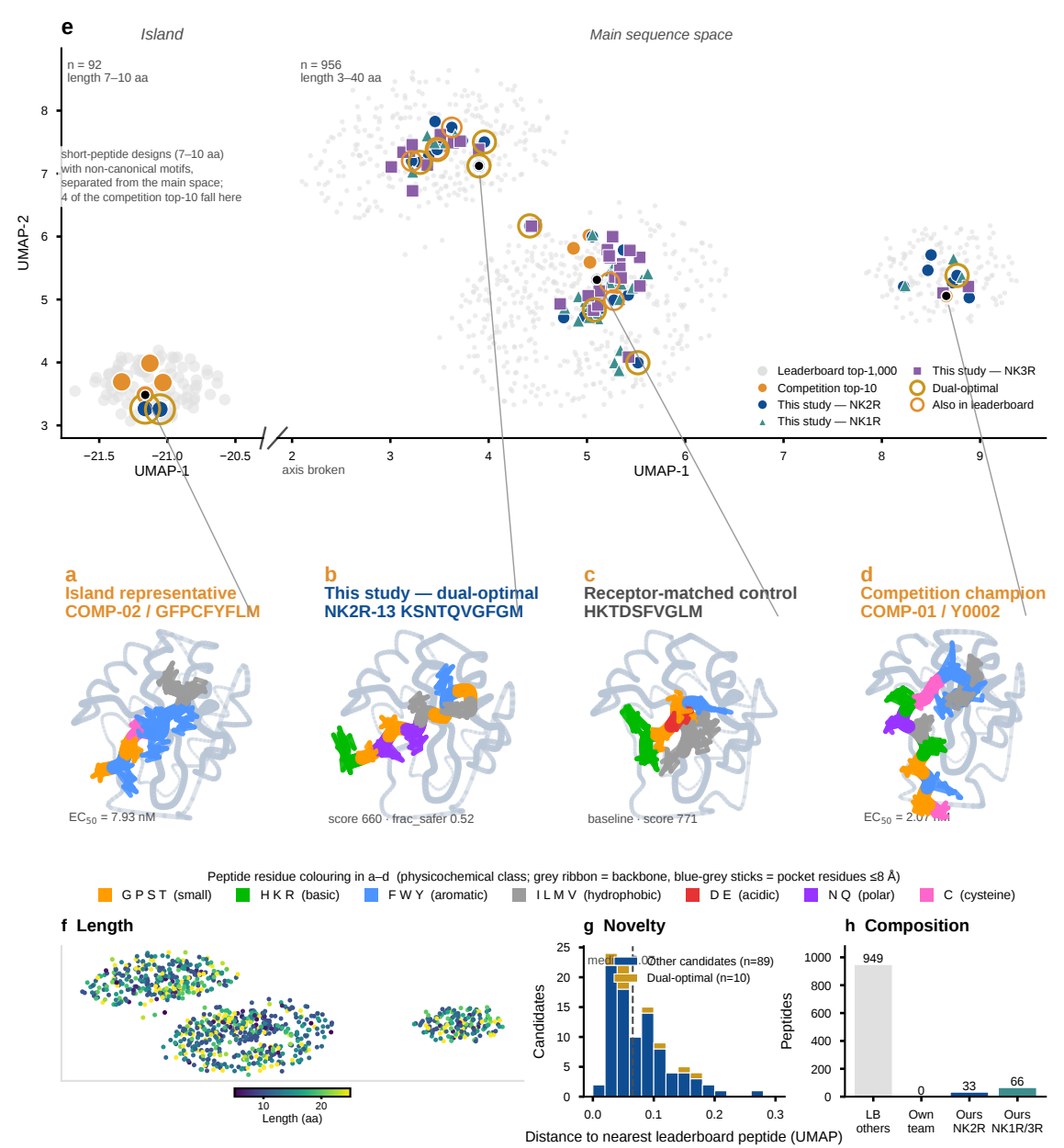

e
Island
Main sequence space
n = 92
length 7–10 aa
n = 956
length 3–40 aa
short-peptide designs (7–10 aa)
with non-canonical motifs,
separated from the main space;
4 of the competition top-10 fall here
UMAP-2
UMAP-1
axis broken
Leaderboard top-1,000
Competition top-10
This study — NK2R
This study — NK1R
This study — NK3R
Dual-optimal
Also in leaderboard
a
Island representative
COMP-02 / GFPCFYFLM
EC50 = 7.93 nM
b
This study — dual-optimal
NK2R-13 KSNTQVGFGM
score 660 · frac_safer 0.52
c
Receptor-matched control
HKTDSFVGLM
baseline · score 771
d
Competition champion
COMP-01 / Y0002
EC50 = 2.07 nM
Peptide residue colouring in a–d (physicochemical class; grey ribbon = backbone, blue-grey sticks = pocket residues ≤8 Å)
G P S T (small)
H K R (basic)
F W Y (aromatic)
I L M V (hydrophobic)
D E (acidic)
N Q (polar)
C (cysteine)
f Length
Length (aa)
g Novelty
median
Other candidates (n=89)
Dual-optimal (n=10)
Candidates
Distance to nearest leaderboard peptide (UMAP)
h Composition
Peptides
949
0
33
66
LB others
Own team
Ours NK2R
Ours NK1R/3R